SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

88/100

Annexure A

Analytical Problem Solving

Code of Conduct:

I Aravindh.R 192312024 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

aravindhR
Signature of the student:

Scanned with OKEN Scanner

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

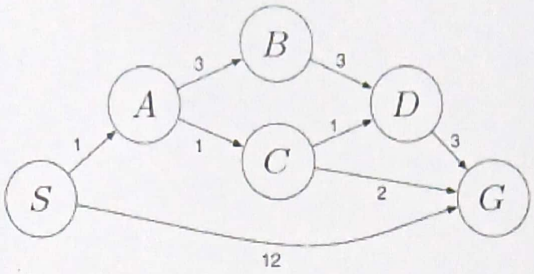
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:

Scanned with OKEN Scanner



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Scanned with OKEN Scanner

SIMATS ENGINEERING
ASSESSMENT TOOL 1 – ANSWER SHEET

Course: Artificial Intelligence | Course Code: CSA17

Analytical Problem Solving

Name: Aravindh R **Register No.: 192312024**

1. Water Jug Problem

Problem understanding: A 4-gallon jug and a 3-gallon jug have no measurement marks. The allowed operations are filling a jug from the pump, emptying a jug onto the ground, and pouring water from one jug into the other. The goal is to obtain exactly 2 gallons in the 4-gallon jug.

Method: Use state-space search. Represent each state as (x, y), where x is the amount in the 4-gallon jug and y is the amount in the 3-gallon jug.

Step	Operation	State (4-gal, 3-gal)	Explanation
0	Start	(0, 0)	Both jugs are empty.
1	Fill 3-gallon jug	(0, 3)	Fill the smaller jug completely.
2	Pour 3-gal into 4-gal	(3, 0)	The 4-gallon jug now contains 3 gallons.
3	Fill 3-gallon jug	(3, 3)	Refill the smaller jug.
4	Pour into 4-gal until full	(4, 2)	Only 1 gallon fits in the 4-gallon jug; 2 gallons remain in the 3-gallon jug.
5	Empty 4-gallon jug	(0, 2)	Empty the 4-gallon jug, keeping 2 gallons in the smaller jug.
6	Pour 3-gal into 4-gal	(2, 0)	Transfer the remaining 2 gallons into the 4-gallon jug.

Result: The 4-gallon jug contains exactly 2 gallons. No external measuring device is used.

2. Mars Rover as an Intelligent Agent

i. Percepts: The rover receives information through cameras, navigation sensors, wheel encoders, temperature and other environmental sensors, and scientific instruments that analyse soil and rock samples. These percepts help it identify obstacles, terrain, landmarks, and sample properties.

ii. Operating environment: Mars is a physical, partially observable, uncertain, sequential, dynamic environment with limited energy, communication delays, and rough terrain. The rover cannot observe every condition at once and must make decisions using incomplete sensor data.

iii. Actions: Move forward or backward, turn, stop, take images, navigate around obstacles, use its robotic arm to collect or handle samples, operate scientific instruments, store data, and transmit results to Earth.

iv. Performance evaluation: Measure safe navigation, distance explored, quality and number of samples analysed, scientific usefulness of collected data, task completion, energy efficiency, equipment safety, and successful communication of results.

v. Suitable architecture: A model-based, goal-based agent with utility-based decision-making is most suitable. A model-based component maintains an internal map and rover status despite partial observability; goal-based

planning selects actions to reach targets and analyse samples; utility-based evaluation balances scientific value, risk, time, and battery power. This hybrid design supports safe and useful autonomous exploration.

3. 8-Queens Problem: Search-Space Formulation

Problem understanding: Place eight queens on an 8 × 8 chessboard so that no two queens share a row, column, or diagonal.

State representation: Represent a state as a list [r1, r2, …, rk], where ri is the row occupied by the queen in column i. This places at most one queen in each column.

Problem components:

- **Initial state:** An empty chessboard, represented by [].
- **State space:** All partial placements of queens in columns 1 through 8, with no attacking pair.
- **Actions:** Place a queen in a safe row of the next column.
- **Transition model:** Append the selected row to the list if it does not conflict with previously placed queens.
- **Goal test:** Eight queens have been placed and no pair attacks each other.
- **Path cost:** A simple formulation assigns cost 1 to each queen placement; every complete solution therefore costs 8.

Constraint test: For queens in columns i and j, their rows must differ ($r_i \neq r_j$), and they must not share a diagonal ($|r_i - r_j| \neq |i - j|$).

Search strategy: Backtracking depth-first search is suitable. Place a queen in the next column, check the row and diagonal constraints, and continue if safe. If no safe row exists, remove the previous queen and try another row.

Example solution (row number for columns 1 to 8): [1, 5, 8, 6, 3, 7, 2, 4]. Each row is used once and no two queens share a diagonal, so this is a valid solution.

4. OLA Cab Booking: Problem-Solving Agent

Suitable agent: A utility-based problem-solving agent is appropriate because the customer chooses among Mini, Micro, Sedan, Shared, Prime, and other cab types according to comfort and possibly fare, travel time, availability, and personal preference. The agent compares available options and selects the one with the highest utility while satisfying the customer's constraints.

Problem formulation:

- **Initial state:** Customer's current location and destination are entered.
- **Actions:** Request available cab types, compare estimates, select a cab, and confirm booking.
- **Goal state:** A suitable cab is booked to take the customer to the destination.
- **Evaluation criteria:** Comfort preference, estimated fare, pickup time, travel time, availability, and any shared-ride preference.

Pseudocode:

```
BEGIN
    INPUT source, destination
    INPUT customer_preferences
    cab_types = [Mini, Micro, Sedan, Shared, Prime]
```

```
    available_cabs = GET_AVAILABLE_CABS(source, destination, cab_types)

    IF available_cabs is empty THEN
        DISPLAY "No cabs available"
        STOP
    END IF

    best_cab = NONE
    highest_utility = -INFINITY

    FOR each cab IN available_cabs DO
        fare = ESTIMATE_FARE(cab, source, destination)
        pickup_time = ESTIMATE_PICKUP_TIME(cab)
        travel_time = ESTIMATE_TRAVEL_TIME(cab, source, destination)
        comfort = MATCH_COMFORT(cab, customer_preferences)
        utility = CALCULATE_UTILITY(comfort, fare, pickup_time, travel_time)

        IF utility > highest_utility THEN
            highest_utility = utility
            best_cab = cab
        END IF
    END FOR

    DISPLAY best_cab and estimated fare
    IF customer confirms THEN
        BOOK(best_cab, source, destination)
        DISPLAY "Cab booked successfully"
    ELSE
        DISPLAY "Booking cancelled"
    END IF
END
```

Interpretation: The selected cab is the best match for the customer's stated preferences among the available options; the cheapest cab is not automatically selected unless cost is the main preference.

5. Uniform Cost Search (UCS) for the Delivery Network

Problem understanding: Find the minimum-total-cost directed path from start node S to goal node G. The edge labels represent route costs (fuel + time). UCS expands the frontier node with the smallest accumulated path cost g(n), rather than the node with the fewest edges.

Edges read from the given graph: S→A = 1, S→G = 12, A→B = 3, A→C = 1, B→D = 3, C→D = 1, C→G = 2, and D→G = 3.

UCS process (frontier entries are ordered by cumulative cost):

Step	Node expanded / event	Frontier after updating (node: cost)
1	Start at S (cost 0)	A:1, G:12
2	Expand A (cost 1)	C:2, B:4, G:12
3	Expand C (cost 2)	D:3, G:4, B:4
4	Expand D (cost 3)	G:4, B:4 (equal-cost order may vary)
5	Remove G at cost 4	Goal reached; stop

Cost comparison:

- S → A → C → G: 1 + 1 + 2 = 4

- S → A → C → D → G: 1 + 1 + 1 + 3 = 6
- S → A → B → D → G: 1 + 3 + 3 + 3 = 10
- S → G: 12

Optimal path: S → A → C → G **Minimum total cost: 4 cost units**

Interpretation: UCS returns the least-cost route because all edge costs are non-negative and the goal is accepted when it is removed from the priority queue with the lowest cumulative cost. The company should use S → A → C → G for the minimum combined travel cost.

Conclusion

These solutions apply state-space representation, intelligent-agent analysis, constraint-based search, utility-based decision-making, and Uniform Cost Search. Each method is selected according to the problem's goal and constraints, and the results are interpreted using the given problem data.